



# NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

MG1002-1 – FINANCIAL MANAGEMENT

(For EE, AD, AM, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five** full questions choosing **Two** full questions from Unit – I & Unit – II each and **One** full question from Unit – III.

*Use of Interest Table is permitted*

## PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What does the Time Value of Money (TVM) concept state?
  - Money grows over time
  - The value of money changes over time
  - Time is valuable
  - Money is the only valuable asset
- Which formula is used to calculate the future value of a present amount?
  - $FV = PV(1+i)^n$
  - $FV = PV \div (1+i)^n$
  - $PV = FV(1+i)^n$
  - $PV = FV \div (1+i)^n$
- By increasing the number of compounding periods in a year, while holding the stated annual interest rate constant, you will.....
  - not change the effective annual rate
  - decrease the effective annual rate
  - increase the effective annual rate
  - There is not enough information to answer the question
- What is the present value of Rs.500 to be received in 2 years with a 5% discount rate?
  - Rs. 452.38
  - Rs. 476.19
  - Rs. 500
  - Rs. 525
- Which of the following best describes the structure of an annuity?
  - A series of payments to be received during a period of time.
  - A series of payments to be received at a common interval during a period of time.
  - A series of equal payments to be received at a common interval during a period of time.
  - The present value of a set of payments to be received during a future period of time.
- How much would a deposit of Rs. 55650 in a bank earning interest @ 15% p.a. grow at the end of 10 years from now?
  - 190368.58
  - 85650.65
  - 210111.76
  - Rs. 2251135.41
- Cash flows accruing to the firms at different time periods are directly comparable.
  - Only in case of Present (PV) Calculations
  - True
  - Only in case of Future Value (FV) calculations
  - False
- The only feasible purpose of financial management is .....
  - Wealth Maximization
  - Sales Maximization
  - Profit Maximization
  - Assets maximization
- Which of the Factor value from the table used when cash flows are uniform to determine the Net Present Value (NPV)
  - Single Payment Present Value Factor (P/F)
  - Uniform Payment Capital Recovery Factor (A/P)
  - Uniform Payment Present Value Factor (P/A)
  - Uniform Payment Sinking Fund Factor (A/F)

10. Profitability index is defined as  
 A) Cash inflows ÷ Payback period  
 B) PV of cash inflows ÷ PV of cash outflow  
 C) ROI ÷ NPV  
 D) Initial investment ÷ Cash inflows
11. Which of the following methods used to determine the number of years required to recover initial investment outlay of the project?  
 A) NPV  
 B) Payback  
 C) ARR  
 D) IRR
12. IRR is compared with  
 A) Cost of capital  
 B) Accounting profit  
 C) Payback period  
 D) Average Cash inflows
13. Which formula is connected to calculate Cost of irredeemable Preference Capital?  
 A)  $K_p = I(1-T)/P_0$   
 B)  $K_p = D \div NP$   
 C)  $K_p = K_e(1-T)$   
 D)  $K_p = NI \div Equity$
14. Working capital is defined as  
 A) Current Liabilities – Current Assets  
 B) Fixed assets – Current Liabilities  
 C) Long terms assets – liabilities  
 D) Current Assets – Current Liabilities
15. Which of the following is NOT a source of long-term capital?  
 A) Equity shares  
 B) Preference shares  
 C) Debentures  
 D) Trade Credit
16. Which one of the following is fixed cost of placing and receiving an order for acquisition of inventory?  
 A) Ordering Cost  
 B) Carrying Cost  
 C) Storage Cost  
 D) None of these
17. The level of inventory at which total cost of inventory consisting of acquisition/ordering and carrying cost is minimal.  
 A) Total Inventory  
 B) Carrying Cost  
 C) Economic Order Quantity (EOQ)  
 D) Work in Process
18. The amount of inventory to be maintained to meet the unforeseen situations and increased lead time to supply the inventory is called as .....  
 A) EOQ  
 B) Safety Stock  
 C) Lead time  
 D) Reorder point
19. Which of the following is not a technique of inventory management?  
 A) Cost-Volume-Profit (CVP) analysis  
 B) JIT  
 C) ABC Analysis  
 D) EOQ
20. A company lists the following items with consumption values:  
 Item – 1: 4000 consumption value  
 Item – 2: 10000 consumption value  
 Item – 3: 1000 consumption value  
 Which item fall under 'A' category in ABC Analysis?  
 A) Item - 3  
 B) Item - 1  
 C) Item - 2  
 D) Item -1 and Item - 3

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I |                                                                                                                                                                                                                                              | Marks | BT* | CO* | PO* |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.       | a) What are the three major questions of long term financial decisions connected to financial operations of the firm covered under the scope of financial management? Explain.                                                               | 06    | L*2 | 1   | 8   |
|          | b) Calculate the future value (FV) using continuous compounding of an investment of Rs.5000 for an interest period of 05 years at 12% rate of interest per annum and the mathematical constant (e) is 2.71828.                               | 04    | L1  | 2   | 1   |
|          | c) A Company paid a dividend of Rs.60 last year. The dividend stream commencing from year one is expected to grow at 10% p.a. for 15 years and then end. If the discount rate is 21%, what is the Present Value (PV) of the expected series? | 06    | L3  | 3   | 2   |

2. a) What is the difference between Nominal Rate of Interest and Effective Rate of Interest? Explain with a suitable numerical example.
- b) If a person borrow Rs. 5 lakh loan from a financial institution for a period of 05 years with 12% rate of interest per year. What is the Equated Monthly Instalment (EMI) to be paid?
- c) If a company paid a dividend of Rs. 10 at the end of last year (i.e., year '0'). The dividend is expected to grow at 8% each year forever (i.e., perpetually). If the discount rate is 10% p.a.; calculate the Present Value (PV) of the dividend.

06 L1 1 8

06 L3 2 2

4 L2 3 2

3. a) What are the major roles and responsibilities of a Financial Manager?
- b) If an investor deposits Rs. 5000 at the beginning of each year for 05 years, calculate the Maturity Value at the end of fifth year if rate of return is 8% per annum.
- c) Calculate the Maturity value (FV) at the end of fifth year if a person invest Rs. 1,00,000 now with an interest rate 12% per annum; compounded:
- Semi-annually
  - Quarterly
  - Monthly

06 L1 1 8

04 L2 2 1

06 L3 3 2

## Unit – II

4. a) What are the features (characteristics) of Capital Budgeting and major types of investment decisions under Capital Budgeting?
- b) Calculate the ARR and select the best proposal for investment from the following:

06 L1 2 9

| Particulars        | Proposal – A | Proposal - B |
|--------------------|--------------|--------------|
| $\sum EBIT(1 - T)$ | Rs. 36000    | Rs. 42000    |
| Cost               | Rs. 50000    | Rs. 60000    |
| Salvage Value      | Rs. 3000     | Rs. 4000     |

Both the proposals work for 05 years.

05 L3 4 2

- c) A Project costs Rs. 50000 with an expected annual income of Rs. 12500 for 05 years. If the discount rate is 8% p.a.; calculate the Profitability Index Value and decide on feasibility of investing in this project.

05 L2 3 1

5. a) The cost of various types of capital of a company is given along with target market proportions. Compute WACC on book-value.

| Source of Funds                | Amount (in Rs.) | Cost of Capital (K) |
|--------------------------------|-----------------|---------------------|
| Debts                          | 240000          | 5.68                |
| Preference shares              | 80000           | 9.33                |
| Equity share<br>(Rs. 100 each) | 400000          | 13.30               |
| Cost of Retained<br>Earnings   | 80000           | 13                  |

- Compute WACC on book-value.
- If, it is in market approach, the price of equity share is Rs. 250 per share instead of Rs. 100 in book value. Calculate the WACC.

08 L3 4 2

- b) A Company proposes to buy a Machine at Rs. 60000 that promises annual savings of Rs. 27000 for next four years. The maintenance expenditure during the 04 years is: zero for the first year and increases by Rs. 2000 each year thereafter. What is the Net Present Value (NPV) of the Machine, if discount rate is 10% p.a.? Is the investment justifiable? 08 L4 4 2
6. a) A firm is considering an expenditure of Rs. 75 lakh for expanding its operations. The relevant information is as follows:  
 Number of existing equity shares – 10 lakh  
 Market Value of the existing shares – Rs. 100  
 Net Earnings – Rs. 100 lakh  
 i. Compute the cost of existing equity share capital  
 ii. Compute new cost of equity capital assuming that the new shares will be issued at a price of Rs. 92 per share and cost of new issues will be Rs. 2 per share. 08 L3 5 2
- b) A workshop manager in a company has to decide select one among the following two machines using the NPV method:

| Particulars                   | Machine – A (Rs.) | Machine – B (Rs.) |
|-------------------------------|-------------------|-------------------|
| First Cost                    | 11000             | 18000             |
| Annual Maintenance Cost (AMC) | 3500              | 3100              |
| Salvage Value                 | 1000              | 2000              |
| Life (in years)               | 06                | 09                |
| Cost of Capital               | 8% p.a.           |                   |

08 L3 3 2

**Unit – III**

7. a) What are inventories? Why companies need to hold or maintain inventories? 04 L1 1 9
- b) Given the details of a product sold:  
 TFC = Rs. 60000; SP per unit = Rs. 5; VC per unit = Rs. 2  
 i. What is BEP?  
 ii. What would be the profit, if the firm sells 30000 units?  
 iii. What would be the BEP, if Rs. 3000 spent on advertising and sale promotion? 06 L3 5 2
- c) A production unit has an expected usage of 50000 units of a certain product during a year. The cost of processing an order is Rs. 20 and the carrying cost per unit is Rs. 0.50 for one year. Lead time on an order is 05 days and the production unit will keep a reserve supply of 02 days usage. Calculate:  
 i. EOQ  
 ii. The Re-order point (Assume 250 production days per year) 06 L4 5 2
8. a) Explain ABC Analysis as an inventory management tool highlighting its advantages and disadvantages. 08 L2 4 9
- b) Given the details:  
 Direct Material cost – Rs. 8 per unit  
 Direct Labour cost – Rs. 5 per unit  
 Fixed overheads – Rs. 24000  
 Variable overheads @ 60% on Direct Labour  
 Selling Price – Rs. 25 per unit  
 Trade Discount – 4%  
 Calculate:  
 i. BEP units  
 ii. Net Profit, if the sales are 20% above the BEP. 08 L3 5 2

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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